



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

COMPUTER SCIENCE

0478/12

Paper 1

May/June 2019

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)	Three from e.g.: – Keyboard – Mouse – Microphone – 2D scanner – 3D scanner – Touchscreen – Webcam // digital camera – Joystick – Trackpad – Sensor – Interactive whiteboard	3
1(b)	Three from e.g.: – Monitor // touchscreen – Inkjet printer – Laser printer – 3D printer – Speaker – Headphones – LED Projector – DLP – 2D cutter – 3D cutter – Actuator	3

Question	Answer	Marks
2(a)	Two from: – It is non-volatile – Can be easily disconnected from the computer – It is not directly accessed by the CPU – Suitable example e.g. CD, DVD, USB flash memory	2
2(b)(i)	One from: – ROM – RAM	1
2(b)(ii)	Two from: – HDD – SSD – Flash memory	2

Question	Answer	Marks
3	1 mark for each correct term, in the correct place: – Data/instructions – Instructions/data (must be the alternative to MP1) – Fetched – RAM – Decoded – Executed	6

Question	Answer	Marks
4(a)	Two from e.g.: – <u>DVD</u> uses red laser/light whereas <u>blu-ray</u> uses blue/violet laser/light – <u>DVD</u> has a smaller (storage) capacity // <u>Blu-ray</u> has a larger (storage) capacity – <u>DVD</u> has two layers (of polycarbonate) whereas <u>Blu-ray</u> disks have a single layer (of polycarbonate) – <u>DVD</u> has a slower transfer rate (of approximately 10 mbps) // <u>Blu-ray</u> has a faster transfer rate (of approximately 36 mbps)	2
4(b)(i)	2 marks for any two correct stages of working, 1 mark for correct answer. – $500 * 300 * 10 // 150\,000 * 10$ – $* 8$ then $\div 8$ (anywhere in the process) – $1\,500\,000 \div 1024 \div 1024 // 1\,500\,000 \div 1\,048\,576$ – $= 1.43 \text{ MB}$	3
4(b)(ii)	Four from: – Rollers are used to move the paper through the printer – Nozzles spray/drop/jet ink onto the paper – Ink jets/print head/nozzles are moved across the paper (to distribute the ink) – Different colour inks are mixed to create required colours – Technology could be piezoelectric – Technology could be thermal bubble – Ink is heated – ... and expands/evaporates into a bubble – Bubble is pushed through the nozzle on to the paper – ... then the bubble collapses – Electrical current is applied to a crystal – ... which makes it vibrate – ... which forces a droplet of ink through the nozzle	4

Question	Answer	Marks
5	– Password protection – Password is released on the release date – Encryption – Encryption key is released on the release date	4

Question	Answer	Marks																		
6(a)	Structure – This is the layout of the web page – e.g. placing an image alongside some text // example of tag, such as <div> Presentation – This is the formatting/style of the web page – e.g. the colour that is applied to some text // example of tag, such as <font-color>	4																		
6(b)	1 mark per each correct row. <table border="1"> <thead> <tr> <th>Statement</th><th>True (✓)</th><th>False (✓)</th></tr> </thead> <tbody> <tr> <td>Cookies can be used to store a customer's credit card details</td><td>✓</td><td></td></tr> <tr> <td>Cookies can be used to track the items a customer has viewed on a website</td><td>✓</td><td></td></tr> <tr> <td>Cookies will corrupt the data on a customer's computer</td><td></td><td>✓</td></tr> <tr> <td>Cookies are downloaded onto a customer's computer</td><td>✓</td><td></td></tr> <tr> <td>Cookies can be deleted from a customer's computer</td><td>✓</td><td></td></tr> </tbody> </table>	Statement	True (✓)	False (✓)	Cookies can be used to store a customer's credit card details	✓		Cookies can be used to track the items a customer has viewed on a website	✓		Cookies will corrupt the data on a customer's computer		✓	Cookies are downloaded onto a customer's computer	✓		Cookies can be deleted from a customer's computer	✓		5
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Question	Answer	Marks
6(c)	– Several/multiple bits are transmitted at a time/simultaneously – Several/multiple wires are used – Data is transmitted in both directions ... – ... at the same time/simultaneously	4
6(d)	One from: – Uniform resource locator – The website's address – User friendly version of the IP address	1
6(e)(i)	Four from: – Designed to deny people access to a website – A large number/numerous requests are sent (to a server) ... – ... all at the same time – The server is unable to respond/struggles to respond to all the requests – The server fails/times out as a result	4
6(e)(ii)	One from: – Proxy server – Firewall	1

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Question	Answer	Marks
7(a)	Six from: – Motion sensor is used – Sensor sends data/signal to microprocessor – Data/Signal is converted from analogue data to digital data (using ADC) – Value is compared to stored value(s) // – If value is outside range/matches ... – ... microprocessor sends signal to switch lights on – ... actuator used to switch light on/off – ... timer is set for 2 minutes – Every time movement is detected the timer is reset – When timer reaches 0/120/times out microprocessor sends signal to switch lights off – Process is continuous	6
7(b)	Three from: – Read only memory – Non-volatile memory // Contents of memory are retained when power is turned off//permanent storage – Primary storage // directly accessed by the CPU – Holds firmware/boot-up instructions/start-up instructions/BIOS – Cannot be written to	3

Question	Answer	Marks
8(a)	1 mark per each correct logic gate, with correct input(s) <pre>graph LR A --- OR1[OR Gate] A --- AND1[AND Gate] B --- OR1 B --- AND2[AND Gate] B --- NOT1[NOT Gate] C --- AND2 C --- OR2[OR Gate] OR1 --- AND2 NOT1 --- OR2 AND1 --- OR3[OR Gate] AND2 --- OR3 OR3 --- X</pre>	6

Question	Answer	Marks																																													
8(b)	4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		0	1	1	0		1	1	1	1		1	4
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Question	Answer	Marks																											
9	1 mark per each correct parity bit: Parity bit <table><tr><td>Register A</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr></table> <table><tr><td>Register B</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> <table><tr><td>Register C</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	Register A	1	0	1	0	0	1	0	1	Register B	1	1	0	0	0	0	0	1	Register C	1	1	0	0	0	0	1	1	3
Register A	1	0	1	0	0	1	0	1																					
Register B	1	1	0	0	0	0	0	1																					
Register C	1	1	0	0	0	0	1	1																					

Question	Answer	Marks
10	Four from: – Electrical field/charge is spread across the screen – Sensors are located around the screen // sensors are used to read the electric field – When finger touches screen, the charge/ is transferred to the user – ... as it is affected by the conductivity of another object – Coordinates of touch determined/calculated/measured	4